

FENG HE

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EDUCATION

M.S. in Neuroscience, University of Science and Technology of China 2023.09 – 2026.06

- Research focus: machine learning, representation learning, and scientific imaging
- Recipient of the **China National Scholarship** (Top 0.1% nationwide), 2026

B.S. in Computer Science, Yangtze University 2019.09 – 2023.06

- GPA: 4.07/5.0; **Ranked 1st** overall among ~300 students (graduate recommendation)

RESEARCH INTERESTS

My research focuses on reliable reasoning and agentic behavior in LLMs, especially how LLMs fail under false premises, hallucinated assumptions, and unreliable verification signals. I develop controlled evaluations, mechanistic analyses, and mitigation methods for robust code generation and agent planning.

SELECTED PUBLICATIONS

Full publication list available on [Google Scholar](#). Total citations: 511 (as of June 2026).

LLM Reliability, Agents, and Reasoning

1. **Feng He**, Hejia Wang, Linghao Meng, Ming Gao, and Qiankun Li. Verification Trap: Understanding Test-Time Selection Failures under False Premises in Code Generation. *ACL ARR 2026 May Submission*.
 - Led the project and identified the Verification Trap, a test-time selection failure mode under false premises, with mechanistic analysis, gold-free prediction, and decoupled-evidence mitigation for code generation.
2. **Feng He**, Linghao Meng, Junyuan Mao, Kun Wang, Liang Lin, Ruixiang Tang, and Qiankun Li. Locked Behavior: Understanding False-Premise Failures in Code Generation. *NeurIPS 2026 Submission*.
 - Led the project and proposed a behavior-level framing of false-premise failures in code generation, including Controlled Behavior Evaluation and behavior-aware analyses of degradation and mitigation.
3. **Feng He**, Zhenting Wang, Dongfang Liu, Ruixiang Tang, and Qiankun Li. Productive LLM Hallucinations: Conditions, Mechanisms, and Benefits. *ECCV 2026 Submission*.
 - Analyzed LLM reasoning under controlled semantic perturbations in scientific domains, characterizing entropy shifts, convergence stability, and mechanism-level factors that govern beneficial hallucinations.

Foundation Models for Scientific and Multimodal AI

4. **Feng He**, Guodong Tan, Qiankun Li, Jun Yu, and Quan Wen. From Pixels to Views: Learning Angular-Aware and Physics-Consistent Representations for Light Field Microscopy. *NeurIPS 2025*.
 - Developed physics-consistent 3D representations for biological light-field microscopy using multi-view angular cues.
5. **Feng He**, Hanlin Li, Xin Ning, and Qiankun Li. BeautyDiffusion: Generative Latent Decomposition for Makeup Transfer via Diffusion Models. *Information Fusion (IF: 18+)*, 2025.
 - Introduced a generative latent decomposition framework enabling disentangled makeup transfer in diffusion models.

HONORS AND AWARDS

Grand Challenge Champion (1st Place)-Micro-Action Recognition ACM MM' 2024

Merit Student; Hanko Scholarship Yangtze University

INTERN EXPERIENCE

Rochester Institute of Technology, Research Collaborator 2025.05 – 2026.01

Collaborating with Prof. Dongfang Liu on research related to hallucination-aware learning and reasoning in large language models. A joint paper is currently under review at *ECCV 2026*.

University of Science and Technology of China, Research Assistant 2023.09 – 2025.05

Conducting research on AI-driven biological image analysis and scientific imaging. Developing novel representation learning methods, resulting in first-author publications at *NeurIPS 2025*.